Today: sidetrack from EB code. N-body sims:

Motivation while Diemer movie plays

↳ PT ⟶ need reference

↳ foundation for physical models (hydro, rad-hydro)

gal form

relaxc

chemistry

↳ cool

Today: Basic ~~N-bo~~ cosmological N-body.

Naive: N particles $\{x_i, p_i\}$

$$\frac{dx_i}{dt} = \frac{p_i}{m} \qquad \frac{dp_i}{dt} = F_i$$

$$F_i \sim \sum_{j \neq i} \frac{1}{|r_i - r_j|^2}$$

$$x_i(t+\Delta t) = x_i(t) + \frac{dx_i}{dt}\Delta t$$

$$p_i(t+\Delta t) = p_i(t) + \frac{dp_i}{dt}\Delta t$$

Need to do this for N particles

F_i needs to sum over N particles

One timestep: $O(N \times N)$ operations... slow.

Instead potential $\nabla^2 \varphi \sim \rho$ $\quad \frac{dp_i}{dt} \sim -\nabla \varphi$

eqn can be solved with FT $\Leftrightarrow$ FFT on mesh $O(\log N_m)$

$\Rightarrow$ particle mesh $\rightarrow$ today.

Expanding b.g. comoving coords

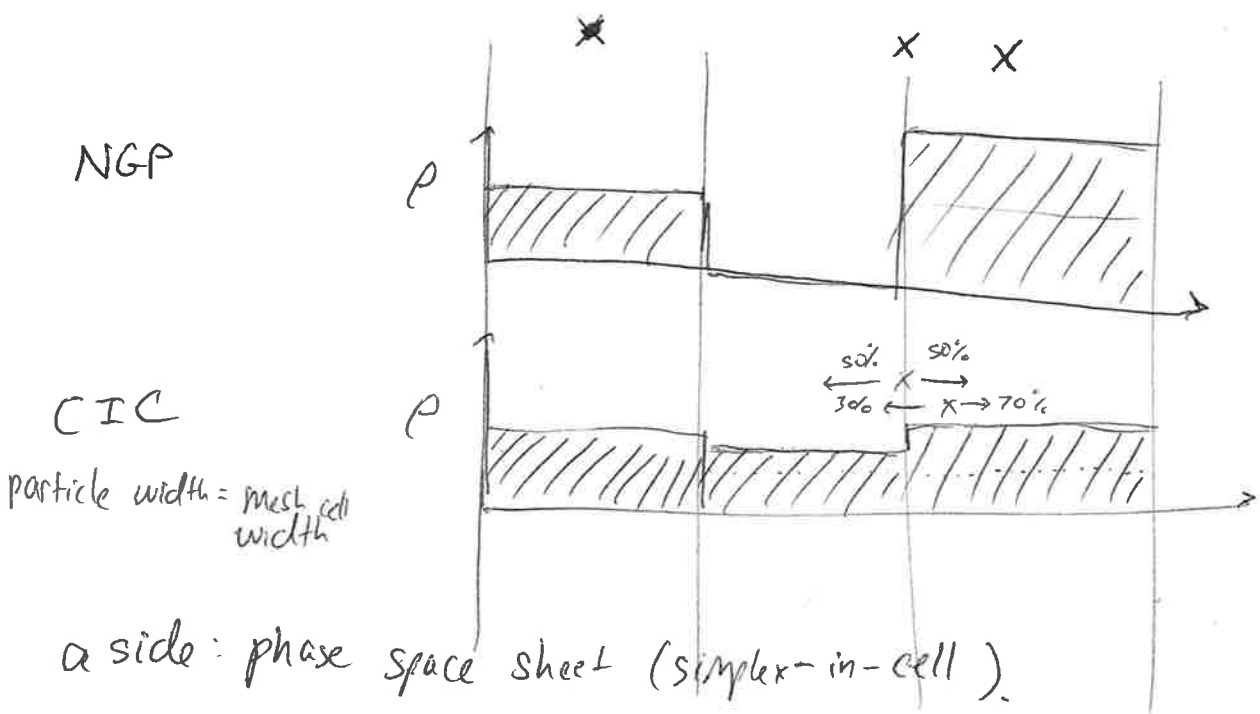
$\rho = \bar{\rho}(1+\delta)$

don't write down $\quad \frac{dx}{da} = \frac{p}{a^{3/2}} \quad \frac{dp}{da} = -\sqrt{a}\,\nabla\varphi \quad \nabla^2\varphi = \frac{\bar{\rho}}{2a}\delta$

1) $\{x_i\} \rightarrow \rho$ on mesh.
2) $\rho \rightarrow \varphi$ FFT
3) time step. repeat

$\{x_i\} \rightarrow \rho$

assume particle has shape.



a side: phase space sheet (simplex-in-cell).

$\rho \to \varphi$ w/ FFT $\quad \rho = \bar{\rho}(1+\delta)$

continuous $\nabla^2 \varphi \sim \delta \to k^2 \varphi \sim \delta \Rightarrow \varphi = \mathcal{F}^{-1}\left[\frac{\delta}{k^2}\right]$

Discrete

index mesh grids $\qquad \nabla \varphi_\mu \sim \varphi_{\mu+1/2} - \varphi_{\mu-1/2}$

$$\nabla^2 \varphi_\mu \sim \varphi_{\mu+1} + \varphi_{\mu-1} - 2\varphi_\mu \sim \delta$$

Linear eq $\Leftrightarrow$ $A\left[\varphi\right] = \left[\delta\right]$

e.g. $\underline{\mu=1}$

$$\varphi_2 + \varphi_0 - 2\varphi_1 \sim \delta_1 \Leftrightarrow [1, -2, 1, 0 \ldots, 0]\left[\varphi\right] = \delta_1$$

$\underline{\mu=2}$

$$\varphi_3 + \varphi_1 - 2\varphi_2 \sim \delta_2 \Leftrightarrow [0\ 1, -2, 1, \ldots 0]\left[\varphi\right] = \delta_2$$

tricky: periodic $\longrightarrow$ $\underline{\mu=0}$

$$\varphi_1 + \varphi_{N_g-1} - 2\varphi_0 \sim \delta_0 \Leftrightarrow [-2, 1, 0 \ldots, 0, 1]\left[\varphi\right] = \delta_0$$

$$A = \begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & 1 & \ddots & \\ & & & \end{bmatrix} \Leftrightarrow A_{\mu\nu} = \begin{cases} -2 & |\mu-\nu| = 0 \\ 1 & |\mu-\nu| = 1 \\ 0 & \text{else.} \end{cases}$$

only depend on $|\mu-\nu| \Rightarrow$ rewrite as convolution.

$$\hat{A} * \varphi = \delta$$

$$(f*g)(x) = \int dy\, f(y-x)\, g(y) \iff (f*g)_i = \sum_j f_{j-i}\, g_j$$

$$\hat{A} = [-2, 1, \ldots, 0, 1] \longrightarrow \delta_1 = (\hat{A} * \varphi)_1 = \sum_{\mu} \hat{A}_{\mu-1}\, \varphi_\mu$$

$$= \underbrace{-2\varphi_1}_{\substack{\mu-1=0 \\ \Rightarrow \mu=1}} + \underbrace{\varphi_2}_{\substack{\mu-1=1 \\ \mu=2}} + \underbrace{\varphi_0}_{\substack{\mu-1=-1 \\ \mu=0}} \quad \checkmark$$

$$\delta \sim \hat{A} * \varphi \xrightarrow{\text{FFT}} \delta(k_\ell) = (\mathcal{F}\hat{A})_\ell\, \varphi(k_\ell)$$

$$k_F = \frac{2\pi}{N_G}, \quad k_\ell = \ell \times \frac{2\pi}{N_g}$$

$$\Rightarrow \varphi(k_\ell) = \underbrace{\left[(\mathcal{F}\hat{A})^{-1}\right]_\ell}_{\text{Green's function}} \delta(k_\ell)$$

$$(\mathcal{F}\hat{A})_\ell = \sum_\mu \hat{A}_\mu\, e^{-ik_\ell \mu}$$

$$= -2e^0 + 1e^{-ik_\ell} + 1e^{-ik_\ell(N_g-1)}$$

$$= -4\sin(k_\ell/2)^2$$

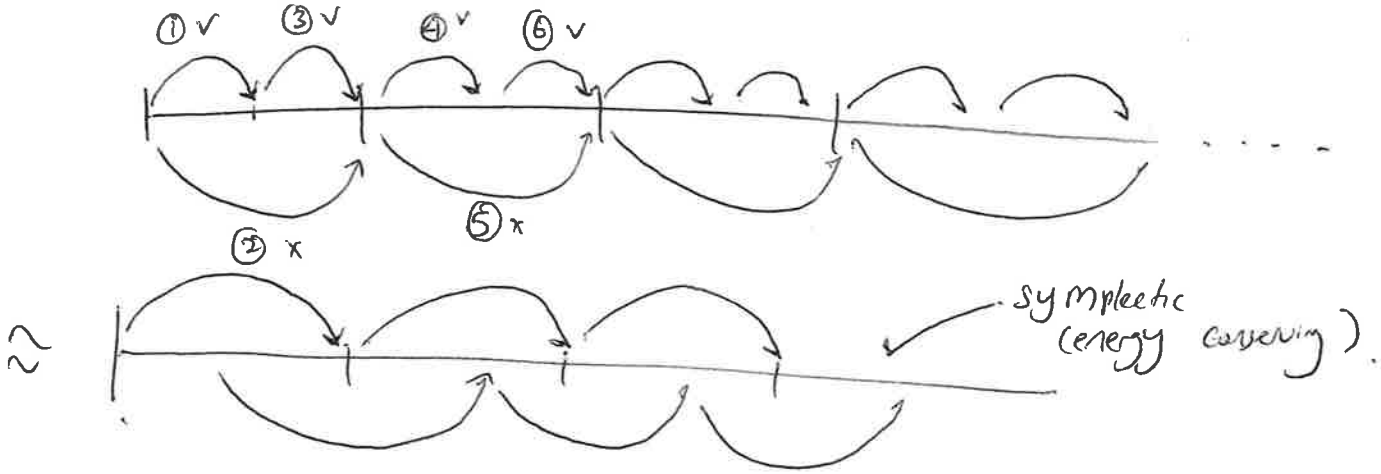
$$\Rightarrow \boxed{\varphi(k_\ell) \sim \frac{\delta(k_\ell)}{\sin(k_\ell/2)^2}}$$

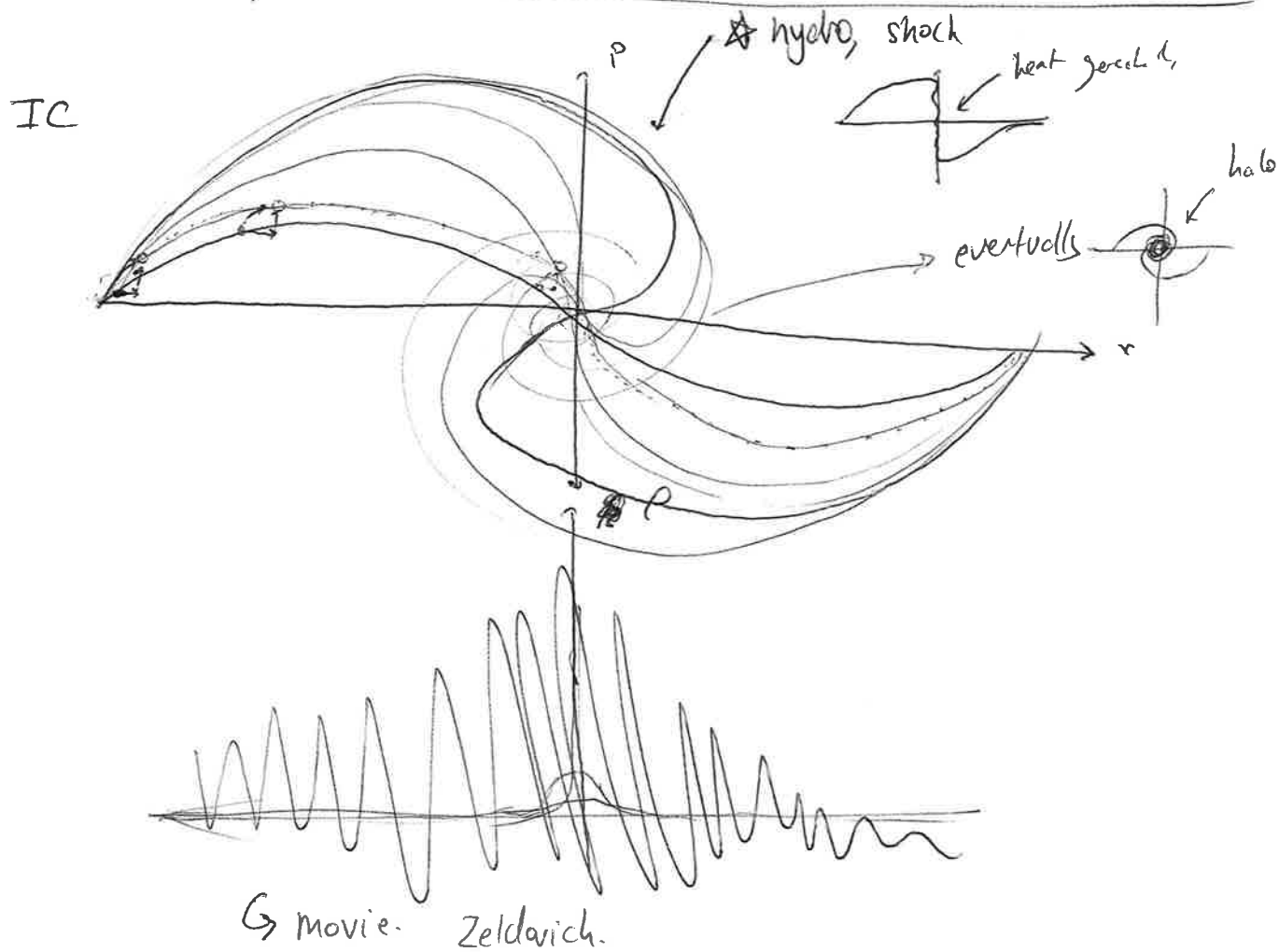
(more detailed derivation in my old pn handout 2)

IFFT $\rightarrow \varphi_\mu$

have $\varphi \to \nabla\varphi$ @ particle (CIC back, see handout)

now can evolve <u>KDK</u> (recipe, see refs for details)





↳ movie. Zeldovich.

↳ aside PIC, LANL plasma